Superstore Sales Performance Analysis

Analyst Memo

Business Question: Which regions, product categories and pricing strategies are driving or destroying profitability across the superstore business?

Data: Superstore transaction dataset, 2014-2017. 9,987 order line items across 4 regions and 3 product categories. Cleaned in Excel, analyzed in MySQL and Excel Pivot Tables.

Key Findings:

1. Technology is the highest-revenue category ($836k), followed closely by office supplies each with a profit margin of 17.4% and 17.02% respectively. Furniture is a loss leader, tables alone produces a $177k loss on $207k in sales, suggesting cost structure or pricing issues in that sub-category.
2. The discount strategy is actively destroying margin. Orders with 40%+ discount average a -77.4% profit margin. 31% of high-discount order lines lose money outright. The business is subsidizing sales volumes with profit. Machines, Binders and Tables receive the highest average discounts and also the most loss-making sub-categories.
3. Sales grew approximately 20% year over year from 2014 to 2017, but Q4 alone accounts for 35% of annual revenue. This concentration creates a cash flow risk. A single weak Q4 would disproportionately impact the year. The pattern is more acute in Technology, where the Q4 spike is nearly 3x the monthly average.

Recommendations:

1. Audit the Furniture pricing and cost structure, particularly Tables. Either raise price, renegotiate supplier cost or reduce the range.

2. The discounting problem is not limited to a handful of products, the majority of sub-categories become loss-making once discount exceeds 20%. Four specific interventions are recommended: cap Binders discount at 30%, cap Machines and Bookcases discounts at 15%, Tables should be placed on a zero-discount policy immediately. The margin structure of this sub-category cannot absorb any promotional pricing and implement a company-wide policy restricting discounts above 20% without senior approval. Only the no discount band generates consistently healthy margins across all sub-categories.
3. Develop a Q1-Q3 promotional strategy in order to reduce the Q4 revenue dependency, especially for technology.